



A CNN-based modular classification scheme for motor imagery using a novel EEG sampling protocol suitable for IoT healthcare systems

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Abstract

The implementation of brain-computer interfaces (BCI) for real-time has become a paramount technology. Implementation of real-time BCI systems requires of methodologies that achieve high performance on classification over general brain signals of different subjects. Therefore, this work presents two simple and efficient methodologies to classify two and four motor imageries. The methodology to classify two motor imageries (MC-TM) includes an analysis of feature extraction methods based on spatial patterns and time–frequency transforms; and a convolutional neural network that preserves the information of the magnitude in the frequency bands of the sensorimotor rhythms (CNN-PIM). Besides, the methodology to classify four motor imageries (MC-FM) includes a modular classification scheme that instances 6 CNN-PIM; a new algorithm that uses the output of the softmax of each CNN-PIM to enhance the performance of the MC-FM methodology; an electroencephalogram sampling protocol that includes a specific procedure for 4 MIs classes; and a new dataset with the brain signals of 15 subjects. The MC-TM methodology achieved an accuracy of $94.44 \pm 02.18\%$ evaluated in BCI Competition IV dataset 2a (BCI-IV-2a), and accuracy of $97.67 \pm 02.06\%$ when evaluated in the EEGdataset. Meanwhile, the MC-FM achieved accuracies of $91.37 \pm 3.29\%$ and $86.48 \pm 4.74\%$ when evaluated in BCI-IV-2a and in the proposed dataset, respectively. These results situate our methodologies in a competitive position in comparison with the state-of-the-art methods. Moreover, the approximate processing time that the MC-FM methodology takes to classify EEG signals is 270 ms. Thus, it is suitable to be implemented in a real-time BCI system.

Keywords Motor imagery · Real-time brain computer interface · Convolutional neural network · Common spatial patterns · Continuous wavelet transform

1 Introduction

As new technologies in communications and networks emerge, the systems that use the Internet to handle data improve. In consequence, different research areas that use the Internet, for example, the Internet of things (IoT) paradigm, have highly increased and even, there are works that seek to regulate its ethical use [1]. Especially, the healthcare IoT area [2] is interesting and complex at the same time. To implement a healthcare IoT system,

different types of data can be used, for instance, getting data through an electrocardiogram (ECG), an electroencephalogram (EEG), or magnetic resonance imaging (MRI).

A recent and popular technology that uses the healthcare IoT paradigm is the brain-computer interface (BCI) [3], in particular, BCI based on motor imagery (MI), e.g., [4]. An MI can be defined as the mental execution of a particular body movement (for example, right-hand movement) without the physical execution of that movement. The register of the brain activity of an MI is commonly done using an EEG. The brain signal involves the mu-wave (7.5–12 Hz), the beta-wave (12–30 Hz), and the sensorimotor rhythms [5, 6].

The BCI systems have been implemented to generate applications in real-time, e.g., [7–9]. In the case of real-time BCI systems based on MI, the input is the brain signal

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from the subjects who performed the MI. The brain signals of the MIs are recorded through an EEG sampling protocol. The EEG sampling protocol is used to give clues to the subject to perform the MI, to find the starting and endings points of the MI signals, and it helps to organize the trials. Thus, the EEG sampling protocol is of extreme importance.

One of the most used datasets to evaluate state-of-the-art works concerning BCI systems based on MI is BCI Competition that uses the traditional Graz paradigm in the EEG sampling protocol [10]. Other datasets have emerged using different EEG sampling protocols such as [11] and [12]. However, the protocols of the datasets [10] and [12], do not indicate how the subject performs the motor visualization and this leads to the generation of infinite combinations of motor visualizations. Thus, it is difficult to ensure that all subjects perform the same motor visualization during the EEG sampling protocol. On the other hand, in [11] a more consistent methodology is proposed in which the motor visualization is limited to a specific finger-movement procedure. This methodology is only used for two MI classes: Left-Hand (LH) and Right-Hand (RH). Therefore, to contribute to the efforts of improving real-time IoT systems, specifically to the implementation of MI-based BCI systems, an EEG sampling protocol is proposed based on [11] including two more MIs: Feet (FT) and Jaw-Clench/Tongue (JC) motions. Besides, two challenges regarding the implementation of real-time MI-based BCI systems are addressed, to generate an efficient and simple general classification method that can be used with the MIs of any subject.

The first challenge is to investigate whether the BCI methodology implemented is suitable for real-time healthcare IoT systems due to the time processing used to classify MIs. This first challenge is important because there are methodologies that use common spatial patterns (CSP) [6, 13–16], others employ time–frequency transformations [17–19] for feature extraction, and others involve a neural network as a classifier [4, 17, 20–27]. However, not all methodologies are suitable for real-time healthcare IoT systems due to their time-processing, amount of data used, complexity, etc.

The second challenge regards the implementation of a classifier that generalizes across the brain activity of all the subjects without being fine-tuned for each subject. In this case, there are novel approaches proposed to solve the challenge of learning to generalize, e.g., multi-task learning [28], meta-learning [29–32], zero-shot learning [33], automated machine learning [34], ensemble learning [35], among others. Nevertheless, our work presents two simple and efficient methodologies to classify two and four MIs.

The main contributions of this paper are summarized in Fig. 1 and described next:

- A Methodology to Classify Two MIs (MC-TM) is used to classify LH and RH MIs of two datasets: *BCI Competition IV dataset 2a* (BCI-IV-2a for short) and *EEGdataset* (EEGd for short). This methodology presents the following characteristics.
- The MC-TM includes an extended analysis of the feature extraction methods CSP, Continuous Wavelet Transform (CWT), Stockwell Transform (ST), CSP + CWT, and CSP + ST presented in [36] which were used with an SVM classifier to classify 2 MIs.
- The extended analysis also includes a modification in the number of rows used in the spatial matrix of the CSP method to improve the features extraction.
- An original Convolutional Neural Network that Preserves the Information of the Magnitude in the frequency bands of the sensorimotor rhythms (CNN-PIM). The architecture of the CNN-PIM is proposed to classify two MIs. In the first part of the paper, the MIs used are LH and RH. Then, other MIs are included: FT and JC motions.
- The performance accuracies of the proposed MC-TM are compared with the accuracies of other state-of-the-art models. The success of the CNN-PIM design in combination with the evaluated feature extraction methods can be seen in the reported results. Especially, our proposed method $CSP_6 + CWT + CNN-PIM$ exceeds the other state-of-the-art models with a $94.44 \pm 02.18\%$ validation accuracy in the BCI-IV-2a dataset, and with a $97.67 \pm 02.06\%$ validation accuracy in the EEGd dataset.
- As a result of the extended analysis of the feature extraction methods and the use of the CNN-PIM, the $CSP_6 + CWT$ method was selected to be used in the Methodology to Classify Four MIs (MC-FM). Subindex 6 in $CSP_6 + CWT$ indicates the number of rows used in the spatial matrix.
- Previous the design of the MC-FM, it was proposed an EEG sampling protocol for MI-based BCI. This sampling protocol enhances the imagination and limits the type of visualization that the subject must generate.
- The proposed EEG sampling protocol includes clues' images and animated images of 4 types of MIs: LH, RH, FT, and JC. This protocol was used to create a new dataset (named as our dataset in this paper) of EEG signals for an MI-based BCI. Our dataset includes the EEG signals of 15 subjects registered through 8 AgCl electrodes.
- Another contribution is a modular classification scheme to classify 4 MIs used in the MC-FM. In the modular classification scheme, the CNN-PIM is instantiated 6 times, one for each pair of combinations of the 4 MIs.

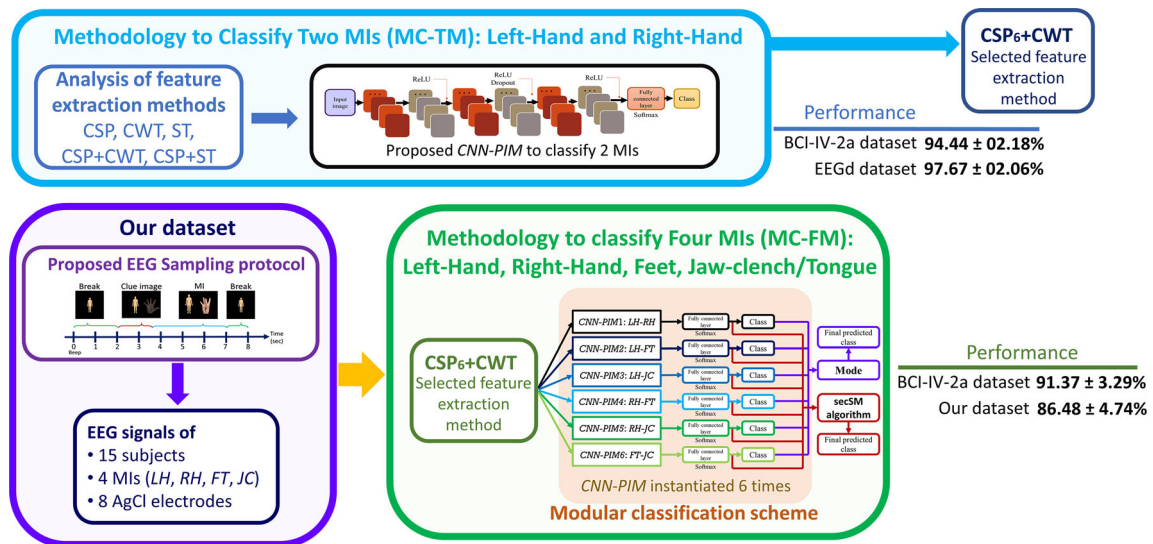


Fig. 1 Diagram of the main contributions of this work

- Another contribution is an algorithm that compares the outputs of the softmax functions of the CNN-PIMs to enhance the performance of the modular classification scheme.
- The success of the use of CSP₆ + CWT + CNN-PIM in the classification of 4 MIs can be seen in the reported results, with $91.37 \pm 03.29\%$ and $86.48 \pm 04.74\%$ validation accuracies when a testing set of the BCI-IV-2a and our dataset were used, respectively. It shows that the performance of our model is competitive regarding the state-of-the-art models.

The remainder of this paper is organized as follows. The description of the MC-TM methodology, which includes the extended analysis of the feature extraction methods, and the design and implementation of the CNN-PIM architecture is presented in Sect. 2. In Sect. 3, the MC-FM methodology is shown, which includes the design and creation of our dataset, the proposed EEG sampling protocol, and the implementation of the modular classification scheme, as well as the results and the analysis of the MC-FM in comparison with state-of-the-art works. Finally, the conclusions are presented in Sect. 4.

2 Methodology to classify two MIs, MC-TM

A. Datasets

The BCI-IV-2a dataset [37] was selected because it is broadly used in other works reported in the literature, and it provides a good form of performance comparison. This dataset includes EEG signals from 4 MI (LH, RH, FT, and JC motions) of 9 subjects. The EEG signals were recorded in 2 sessions with 288 trials each session (72 trials for each

MI for each session). The sampling frequency, F_s , used was 250 Hz. A 0.5–100 Hz bandpass filter was applied.

The EEGd dataset was obtained from [11]. It was selected because it includes many subjects (52). It is one of the most up-to-date MI-datasets (2017) and includes EEG signals from 2 MI (LH, and RH). The EEG signals were recorded in one session with 100–120 trials for each MI, and with a sampling frequency of 512 Hz.

In the BCI-IV-2a and EEGd datasets, the number of trials for each MI for all sessions is defined by p_{dataset} , where $\text{dataset} = \{\text{BCI}, \text{EEGd}\}$, i.e., $p_{\text{BCI}} = 144$ and $100 \leq p_{\text{EEGd}} \leq 120$. Both datasets registered EEG signals with the Ag/AgCl electrodes. The number of electrodes (channels) nCh_{dataset} is $nCh_{\text{BCI}} = 22$ and $nCh_{\text{EEGd}} = 64$. For mathematical notation simplicity, in future equations, the constants p , nCh , and F_s will not include the index of the dataset.

B. EEG signal processing

The EEG signal processing in this work includes pre-processing, epochs extraction, and data augmentation stages, as it is done in [36].

(1) Preprocessing

Let $X_{\text{raw}}^{\text{Ch}}(n) \in \mathbb{R}^{nCh \times N}$ be the discrete signal which contains N raw data of and EEG channel Ch from each subject Φ . As this section addresses the methodology for two MI, in the case of the BCI-IV-2a dataset only the LH and RH MIs signals were considered.

The signal $X_{\text{raw}}^{\text{Ch}}(n)$ was filtered with a fifth-order bandpass IIR Butterworth digital filter, with cut frequencies corresponding to the frequencies of the waves mu and beta, 7.5–30 Hz. This filter allows a monotonic response in the

band pass up to the cut frequency, and the magnitude of the raw data is not modified. $X_{\text{filt}}^{\text{Ch}}(n) \in \mathbb{R}^{n \times \text{Ch} \times N}$ is the filtered signal of each Channel Ch for subject Φ for dataset.

(2) Epochs extraction

This stage consists of the extraction of p epochs, $ep = 1, 2, \dots, p$, for each MI class to reduce the data dimensionality, and to extract the relevant information of $X_{\text{filt}}^{\text{Ch}}(n)$,

$$x_{\Phi, ep}^{\text{Ch}}(k) = X_{\text{filt}}^{\text{Ch}}(q_{ep} + k + \text{inc}); x_{\Phi, ep}(k) \in \mathbb{R}^{n \times \text{Ch} \times 2Fs \times p} \quad (1)$$

where $k = 1, 2, \dots, 2Fs$ and it is the data index in the epoch ep of the channel Ch and q_{ep} refers to the initial sample of the MI in the epoch ep . Therefore, there are p epochs of two seconds ($2Fs$ samples) of each channel Ch of each MI. At the first instance $\text{inc} = 45$ (number of samples that correspond to 180 ms at $Fs = 250$ Hz), because it is intended to obtain an epoch going from 180 ms after the beginning of the MI ($q_{ep} + \text{inc}$) to 2180 ms ($q_{ep} + 2Fs + \text{inc}$). This interval is selected because an MI presents an event-related desynchronization or ERD with a duration of 1500 ms to 2000 ms [38], and thus assuring to obtain adequate information to analyze the type of MI.

(3) Data augmentation

In order to represent better each MI class, it is necessary to increment the number of samples in the EEG signal under analysis [24]. Data augmentation is achieved by overlapping windows of 80 ms in 80 ms that involve the MI event under screening. In this way, the most important information with only a few modifications is preserved [39]. Therefore, inc takes different values in (6)

$$\text{inc} = 45 + (20)(l); l = 0, 1, 2, \dots, 9 \quad (2)$$

i.e., inc takes 10 different values from 45 (180 ms) to 225 (900 ms), by the increments of 20 samples every 80 ms. This results in the generation of 10 epochs per trial instead of one epoch per trial. The first epoch will cover the interval in the MI event from 180 to 2180 ms and the last one from 900 to 2900 ms. These data were divided into two sets: $8p$ epochs of each MI that represent 80% of the samples were employed for training and the other 20% for validation.

C. Feature extraction

This section presents the five methods used for feature extraction that were tested in this work: CSP, CWT, ST, and the following combinations CSP + CWT and CSP + ST. In the next paragraphs, each feature extraction method is explained and the classifier to recognize right and left-hand MIs using the previous features will be analyzed.

(1) Method 1: common spatial patterns, CSP

By using CSP we try to do a linear transformation that projects a multichannel EEG signal into a subspace of lower dimension. This process allows to use information from both classes and eliminates redundant information. In this methodology of two MIs, the CSP are generated by considering right and left-hand MIs.

As it is done in [36], the spatial filter matrix is defined by $W_{\text{CSP}} \in \mathbb{R}^{n \times \text{Ch}}$, where the rows correspond to the combined information of both classes of each channel employed to obtain the EEG signal. This information is arranged in descending order where the first row has the highest variance. For mathematical notation simplicity, the subject Φ will be omitted in the future equations. It is important to say that each subject has his/her spatial filter matrix. The transformation of each epoch is done by using the first row and the $n\text{Ch}$ row of the spatial filter

$$\mathbf{Z}_{\text{CSP}}^{\text{ep, row}}(k) = [W_{\text{CSP}}^{\text{row}}]^T [x_{ep}(k)]; \mathbf{Z}_{\text{CSP}} \in \mathbb{R}^{2 \times 2Fs \times 8p} \quad (3)$$

where row represents each row used in the transformation, and $8p$ corresponds to the number of epochs used in the training stage. This produces a reduction of the data dimension, and it concatenates the two rows with higher variances to obtain the matrices $\mathbf{Z}_{\text{CSP}}^{\text{ep, row}}(k)$.

Since the architecture of the CNN-PIM was inspired by [24] (more details of the CNN-PIM architecture are given in Sect. 2D), $\mathbf{Z}_{\text{CSP}}^{\text{ep, row}}(k)$ is resized through a bicubic interpolation method. In the bicubic interpolation method, the value of the output pixel is a weighted average of the pixels in the nearest 4×4 neighborhood [40]. In this work, the bicubic interpolation operator is defined by BI. Applying the bicubic interpolation to the matrix $\mathbf{Z}_{\text{CSP}}$ results in

$$\text{BI}[\mathbf{Z}_{\text{CSP}}] \rightarrow I_{\text{CSP}} \in \mathbb{R}^{22 \times 100 \times 8p} \quad (4)$$

where I_{CSP} contains the input images for the CNN-PIM, with information of CSP method.

(2) Method 2: continuous Wavelet transform, CWT

This method involves the continuous Wavelet transform using the Morlet wavelet, and is applied on the channels $\text{Ch} = \{C3, C4\}$ of each one of the $8p$ epochs

$$\mathbf{Z}_{\text{CWT}}^{\text{ep, Ch, sc}}(k) = \text{abs}\left(\text{CWT}\left\{x_{ep}^{\text{Ch}}(k)\right\}\right) \quad (5)$$

where $\mathbf{Z}_{\text{CWT}}(k) \in \mathbb{R}^{SC \times 2Fs \times 2 \times 8p}$. The output frequency scale of the CWT (represented in logarithmic scale base 2) is discretized using 10 voices per octave. Both the minimum and maximum values of the output frequency scale are determined automatically as a function of the energy distribution of the CWT [41]. The index $sc = \{1, 2, \dots, SC\}$ is used to point out each value included in the output frequency scale from 7.5 to 28 Hz (sensorimotor rhythms). Then, the resulting matrices of the CWT are used by each channel $\{C3, C4\}$ to compute the matrix

$$\mathbf{Q}_{\text{CWT}}^{ep}(i, k) = \begin{cases} \mathbf{Z}_{\text{CWT}}^{ep, C4, (i-SC)}(k) & i > SC \\ \mathbf{Z}_{\text{CWT}}^{ep, C3, i}(k) & i \leq SC \end{cases}; i = 1, 2, \dots, 2SC; \mathbf{Q}_{\text{CWT}} \in \mathbb{R}^{2SC \times 2Fs \times 8p} \quad (6)$$

here, each row of each one of the matrices $\mathbf{Z}_{\text{CWT}}^{ep, Ch, sc}(k)$ corresponds to a specific value of the output frequency scale. The last step is to resize the matrix

$$\text{BI}[\mathbf{Q}_{\text{CWT}}] \rightarrow \mathbf{I}_{\text{CWT}} \in \mathbb{R}^{22 \times 100 \times 8p} \quad (7)$$

where $\mathbf{I}_{\text{CWT}}$ contains the input images for the CNN-PIM, with information of CWT method.

(3) Method 3: Stockwell transform, ST

This third method uses the Stockwell transform, over the channels C3 and C4 of each one of the epochs, employing variable gaussian windows

$$\mathbf{Z}_{\text{ST}}^{ep, Ch, f}(k) = \text{abs}\left(\text{ST}\left\{x_{ep}^{Ch}(k)\right\}\right) \quad (8)$$

where $\mathbf{Z}_{\text{ST}} \in \mathbb{R}^{F \times 2Fs \times 2 \times 8p}$ and $f = \{1, 2, \dots, F\}$ is an index of frequencies (from 1 to F_s Hz) obtained from the transformation, represented with a resolution of 1 Hz, selecting the band 8–28 Hz, that contains the sensorimotor rhythms. Unlike in the CWT method, in the ST transform the range of frequencies of 8–28 Hz was used instead of the band 7.5–28 Hz, because the ST has a resolution of 1 Hz, and the resolution of the CWT is a function of the output frequency scale of the transform.

Similar to the CWT method, the resulting matrices of the ST are used by each channel {C3, C4} to compute the matrix

$$\mathbf{Q}_{\text{ST}}^{ep}(i, k) = \begin{cases} \mathbf{Z}_{\text{ST}}^{ep, C4, (i-F)}(k) & i > F \\ \mathbf{Z}_{\text{ST}}^{ep, C3, i}(k) & i \leq F \end{cases}; i = 1, 2, \dots, 2F; \mathbf{Q}_{\text{ST}} \in \mathbb{R}^{2F \times 2Fs \times 8p} \quad (9)$$

here, each row of each one of the matrices $\mathbf{Z}_{\text{ST}}^{ep, Ch, f}(k)$ of the ST corresponds to a frequency. Then, the bicubic interpolation operator is applied

$$\text{BI}[\mathbf{Q}_{\text{ST}}] \rightarrow \mathbf{I}_{\text{ST}} \in \mathbb{R}^{22 \times 100 \times 8p} \quad (10)$$

where $\mathbf{I}_{\text{ST}}$ contains the input images for the CNN-PIM, with information of ST method.

(4) Method 4: CSP + CWT

This method involves the combination of the features extracted with the methods CSP, and the CWT. The CSP was used to automatically select the two rows of the filter $\mathbf{W}_{\text{CSP}}$ that include information with the highest variance of both classes. CWT was used to process this information. Therefore, the CWT was obtained from the matrix yielded by the CSP

$$\mathbf{Z}_{\text{CSP_CWT}}^{ep, \text{row}, sc}(k) = \text{abs}(\text{CWT}\{\mathbf{Z}_{\text{CSP}}^{ep, \text{row}}(k)\}) \quad (11)$$

where $\mathbf{Z}_{\text{CSP_CWT}} \in \mathbb{R}^{SC \times 2Fs \times 2 \times 8p}$ is the absolute value obtained from the information of both classes of the two selected rows of the spatial matrix of the CSP. Then, the resulting matrices of the CSP + CWT method are used to compute the matrix

$$\mathbf{Q}_{\text{CSP_CWT}}^{ep}(i, k) = \begin{cases} \mathbf{Z}_{\text{CSP_CWT}}^{ep, 2, (i-SC)}(k) & i > SC \\ \mathbf{Z}_{\text{CSP_CWT}}^{ep, 1, i}(k) & i \leq SC \end{cases}; i = 1, 2, \dots, 2SC; \mathbf{Q}_{\text{CSP_CWT}} \in \mathbb{R}^{2SC \times 2Fs \times 8p} \quad (12)$$

and the bicubic interpolation method is applied

$$\text{BI}[\mathbf{Q}_{\text{CSP_CWT}}] \rightarrow \mathbf{I}_{\text{CSP_CWT}} \in \mathbb{R}^{22 \times 100 \times 8p} \quad (13)$$

where $\mathbf{I}_{\text{CSP_CWT}}$ contains the input images for the CNN-PIM, with information of CSP + CWT method.

An example of an input image for the CNN-PIM of an LH MI is shown in Fig. 2a, where the matrices of the CSP + CWT were concatenated, and the bicubic interpolation method was applied. An example of an RH MI is shown in Fig. 2b.

(5) Method 5: CSP + ST

The CSP filters select the best channels, and the ST is used to analyze and select the time–frequency features

$$\mathbf{Z}_{\text{CSP_ST}}^{ep, \text{row}, f}(k) = \text{abs}(\text{ST}\{\mathbf{Z}_{\text{CSP}}^{ep, \text{row}}(k)\}) \quad (14)$$

here $\mathbf{Z}_{\text{CSP_ST}} \in \mathbb{R}^{F \times 2Fs \times 2 \times 8p}$ is the absolute value obtained from the information of both classes of the two selected rows of the spatial matrix of the CSP. Then, the resulting matrices of the CSP + ST method are used to compute the matrix

$$\mathbf{Q}_{\text{CSP_ST}}^{ep}(i, k) = \begin{cases} \mathbf{Z}_{\text{CSP_ST}}^{ep, 2, (i-F)}(k) & i > F \\ \mathbf{Z}_{\text{CSP_ST}}^{ep, 1, i}(k) & i \leq F \end{cases}; i = 1, 2, \dots, 2F; \mathbf{Q}_{\text{CSP_ST}} \in \mathbb{R}^{2F \times 2Fs \times 8p} \quad (15)$$

and the bicubic interpolation method is applied

$$\text{BI}[\mathbf{Q}_{\text{CSP_ST}}] \rightarrow \mathbf{I}_{\text{CSP_ST}} \in \mathbb{R}^{22 \times 100 \times 8p} \quad (16)$$

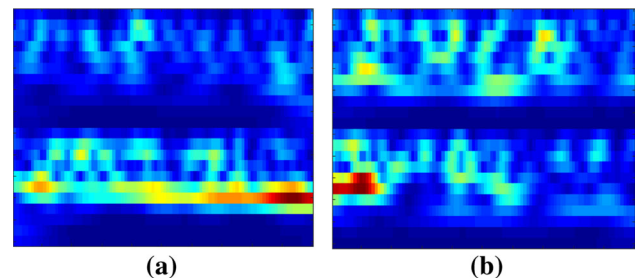


Fig. 2 Examples of CSP + CWT image of a an LH MI b an RH MI

where I_{CSP_ST} contains the input images for the CNN-PIM, with information of CSP + ST method.

D. Two-MIs Classifier

The CNN-PIM design is based on [24] where the authors propose the use of the Power Spectral Density (PSD) and the Welch Transform as feature extractor methods, in order to classify four MIs using the BCI-IV-2a dataset. In [24] it is mentioned that the lack of convolutional layers can lead to poor classification performance, so they propose a CNN with 9 convolutional layers. However, this assumption is not always fulfilled. In this work it is proposed a CNN architecture with only 3 convolutional layers, applying a dropout layer after the second convolutional layer to improve the generalization of the learning in the CNN. The use of the 3 convolutional layers in the CNN-PIM model was obtained methodological by varying the number of convolutional layers and registering its performance. The highest performance was attained with 3 convolutional layers.

The principle that is followed to reduce the number of convolutional layers is the trade-off between layers and the number of filters used in each layer. For example, in this work, the number of layers is reduced meanwhile the number of filters is augmented. This was done because the more hidden convolutional layers a CNN has, the features obtained will be more specifics [42] and in this particular case it could be prejudicial to the performance of the MC-TM methodology, and it is explained next. The input of the CNN-PIM is a matrix with time–frequency information, and it is important to preserve the information of the magnitude in the frequency bands of the sensorimotor rhythms (information regarding the frequencies of 7.5–30 Hz), i.e., if the features are very specific, the representation of the frequency band of interest (sensorimotor rhythms) will be reduced and the information regarding MIs will be lost. The design of the proposed CNN Preserves the Information of the Magnitude in the frequency bands of the sensorimotor rhythms, and therefore the proposed CNN in this paper is named CNN-PIM. In such a way that it is expected to achieve better results in the classification implementing a CNN-PIM architecture with more convolutional filters that allow obtaining more

general information from the time–frequency input matrix. Also, using fewer convolutional layers, the features obtained by the CNN preserve the information included in the band of 7.5–30 Hz.

The architecture of the CNN-PIM proposed is shown in Fig. 3. Each convolutional layer has 128 convolutional filters of 3×3 . At the output of each convolutional layer there is a 2×2 max-pooling layer with *stride* = 1, and a ReLU activation function. As in [24], the weights were initialized using Glorot. At the end of the third convolutional layer, there is a fully-connected layer and *Softmax* function. The Adam optimizer [43] was used.

The learning rate and other hyperparameters used in the training of the CNN-PIM (Table 1) were obtained based on previously reported information and then fine-tuned. For instance, we based the choice of the mini-batch size using the information provided in the Deep Learning book by Ian Goodfellow, et al. [44]. In [44] it is mentioned that a typical mini-batch size is a power of 2, in the interval 32 to 256. The value used was 128.

E. Results of the two-MIs classification

The performances using the proposed feature extraction methods and the CNN-PIM as the classifier were obtained

Table 1 Hyperparameters used in the training of the CNN-PIM

Hyperparameter	Value
Optimizer	ADAM
Gradient decay factor	0.1900
Squared gradient decay factor	0.9000
Epsilon	1.0e-8
Initial learn rate	5.0e-4
L ₂ regularization	1.0e-4
Learn rate drop factor	0.99
Learn rate drop period	3
Learn rate schedule	Piecewise
Max training-epochs	60
Mini batch size	128
Shuffle	Once
Execution environment	GPU

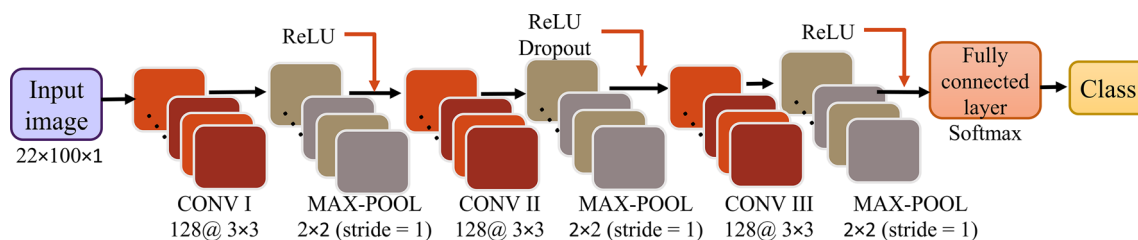


Fig. 3 Architecture of the proposed CNN-PIM

by using the provided *Ground Truth* in the datasets for the training and validation sets. Using the training set of the BCI-IV-2a dataset, the average accuracy for all subjects for each feature extraction method was $84.70 \pm 12.45\%$ for CSP_2 , $86.41 \pm 4.78\%$ for CWT, $76.12 \pm 8.01\%$ for ST, $87.69 \pm 9.66\%$ for $CSP_2 + CWT$, $84.99 \pm 10.02\%$ for $CSP_2 + ST$. Using the validation set of the BCI-IV-2a dataset a more complete analysis per subject is proposed in Table 2. Table 2 uses a color code. The color code represents a scale that passes between the green-yellow-red colors. The green color in this scale indicates the best performance, the yellow color indicates an average performance, and the red color corresponds to the worst performance. In this case, the color code was implemented by subject. The hyperparameters of the CNN-PIM were fixed in all trainings with the values of Table 1. CSP_2 indicates the use of two rows of the spatial matrix.

It can be observed in Table 2 that the standard deviation, STD values are in the interval of 6.78% to 13.72%. These values of STD are significantly reduced in our proposed method $CSP_6 + CWT$ explained in the next section. The range of the variances is because the architecture of the CNN-PIM is a general model for all the subjects that are used in this paper and there are subjects whose MIs are easier to classify than others using the MC-TM methodology. For example, the performance classification of the MIs using $CSP_2 + CWT$ of subject 02 (65.80%) of the BCI-IV-2a dataset was not as good as the performance of subject 08 (98.61%). Nevertheless, the proposed MC-TM methodology shows performances over 80% of correct classification, especially when using the combined feature extraction method $CSP_2 + CWT$.

For the training and validation sets of the EEGd dataset the results are presented next. Using the training set, the average accuracy for all subjects for each feature extraction method was $85.87 \pm 6.68\%$ for CSP_2 , $89.14 \pm 8.12\%$ for CWT, $91.83 \pm 6.49\%$ for ST, $94.57 \pm 4.2\%$ for $CSP_2 + CWT$, $95.3 \pm 3.66\%$ for $CSP_2 + ST$. Using the validation set, a more complete analysis per subject is proposed in Table 3. The performances shown in Table 3 can be directly compared with the results in [11], which is the paper where the EEGd dataset is proposed. In [11] the average performance reported over all subjects is $67.46 \pm 13.17\%$. The best validation performance attained by our MC-TM methodology is $91.82 \pm 3.96\%$ using $CSP_2 + ST$ and the CNN-PIM.

One of the advantages of using CNNs is to avoid hand-craft feature extraction. However, there are works like [45] where it is shown that the CNNs can be easily fooled using distorted data, producing that the CNNs classify images incorrectly. Thus, it is important to consider the data-centric approach when implementing deep learning models like CNNs. Considering the data-centric approach, the data

Table 2 Accuracies per subject of the proposed MC-TM methodology using the CNN-PIM with the validation set of BCI-IV-2A dataset

S	CSP_2	CWT	ST	$CSP_2 + CWT$	$CSP_2 + ST$
01	93.58%	78.65%	78.82%	87.67%	86.46%
02	54.17%	71.35%	65.28%	65.80%	69.27%
03	97.40%	83.16%	83.85%	94.27%	95.31%
04	75.52%	69.79%	61.81%	82.81%	65.80%
05	70.31%	76.39%	68.23%	72.74%	76.91%
06	78.13%	71.70%	64.93%	76.56%	69.62%
07	76.56%	73.61%	68.40%	69.62%	64.76%
08	94.79%	72.40%	70.83%	98.61%	94.79%
09	82.29%	90.63%	65.80%	88.19%	89.76%
AVG	80.30%	76.41%	69.77%	81.81%	79.19%
STD	13.72%	6.78%	7.15%	11.34%	12.50%

must be the most adequate possible to solve a specific task. In this work, the task consists in discriminate MIs signals of different subjects, where the time–frequency information of MIs signals is treated as an image. In consequence, to obtain the most adequate images to solve the task of classifying MIs using a CNN, the proposed feature extraction methods are used and analyzed.

From the previous Tables 2, 3, it can be noticed that the combination of spatial patterns with time–frequency transforms produces the best performances. Therefore, it is proposed to augment the number of rows used in the spatial matrix to improve the $CSP_2 + CWT$ and $CSP_2 + ST$ feature extraction methods. The hypothesis here is that if the number of rows used in CSP augments, the discriminant capacity increases too, and, in consequence, the performance will be better. The next section presents the results of using 4 and 6 rows in the spatial matrix of CSP.

(1) CSP_4 and CSP_6

To extract features of the input images through CSP involving 4 and 6 rows, the procedure described in Sect. 2C1 is modified to $row = \{1, 2, nCh - 1, nCh\}$ to include CSP_4 , and $row = \{1, 2, 3, nCh - 2, nCh - 1, nCh\}$ for CSP_6 . The $Q_{CSP_CWT}^{ep}$ and $Q_{CSP_ST}^{ep}$ matrices, and the bicubic interpolation to generate 22×100 images are computed as in Sect. 2C4 and Sect. 2C5, for CWT and ST, respectively. The CNN-PIM architecture (Fig. 3) is the same, and the hyperparameters maintain the values shown in Table 1.

The results of the CNN-PIM using CSP_4 and CSP_6 with CWT or with ST as feature extraction methods for the training and validation sets of the BCI-IV-2a dataset are presented next. Using the training set, the average accuracy for all subjects for each feature extraction method was $92.16 \pm 4.59\%$ for $CSP_4 + CWT$, $89.23 \pm 5.08\%$ for $CSP_4 + ST$, $98.08 \pm 1.4\%$ for $CSP_6 + CWT$,

$95.54 \pm 2.99\%$ for $CSP_6 + ST$. Using the validation set, a more complete analysis per subject is shown in Table 4.

The results of the CNN-PIM using CSP involving 4 and 6 rows of the spatial matrix of CSP for the training and validation sets of the EEGd dataset are presented next. Using the training set, the average accuracy for all subjects for each feature extraction method was $98.56 \pm 1.88\%$ for $CSP_4 + CWT$, $95.81 \pm 4.8\%$ for $CSP_4 + ST$, $98.83 \pm 1.4\%$ for $CSP_6 + CWT$, $98.65 \pm 1.93\%$ for $CSP_6 + ST$. A more complete analysis per subject, accuracy, and the STD, using the validation set is shown in Table 5.

The hypothesis of increasing the number of rows used in CSP to augment the discriminant capacity was fulfilled. Using 4 and 6 rows in the spatial matrix of CSP enhanced the performances of the CNN-PIM in both datasets: BCI-IV-2a and EEGd. Therefore, to complement the analysis of the reported feature extraction methods, Table 6 summarizes the accuracy performances of other methods presented in the literature. Table 6 indicates the feature extraction method, the classifier, the number of motor imageries, the number of test subjects, the dataset evaluated, and the reference and the year of each work.

Table 6 shows the results for BCI-IV and EEGd datasets. The reported methods in the literature present accuracies around 80% with an STD around 11% in the BCI-IV dataset. The best reported method is AR-CSP + SRSF-FasAr [47], with an accuracy of $85.49 \pm 11.64\%$. On the other hand, using the first 5 feature extraction methods reported in our analysis CSP_2 , CWT, ST, $CSP_2 + CWT$, and $CSP_2 + ST$ yield accuracies, using CNN-PIM as the classifier, near to 80% (except ST with 69.77%) with similar values of STD as the reported in the literature. However, when our enhanced feature extraction methods $CSP_6 + ST + CNN-PIM$ and $CSP_6 + CWT + CNN-PIM$ were used, the performance accuracy for the classification of two motor imageries improved significantly to $90.88 \pm 05.14\%$ and $94.44 \pm 02.18\%$, respectively. In addition, the STD value decreased significantly. In the EEGd dataset the method with the highest accuracy is VMD + STFT + DeepConvNet [46] with $88.51 \pm 10.64\%$. The first 5 feature extraction methods reported in our paper CSP_2 , CWT, ST, $CSP_2 + CWT$, and $CSP_2 + ST$ generated accuracies of 80% to 91% using the CNN-PIM classifier. This result means that even when just 2 rows of the spatial matrix of the CSP method in combination with the time–frequency transform methods ($CSP_2 + CWT$, and $CSP_2 + ST$) the performances of our methods, overcome the results reported in the literature. Besides, using the features $CSP_6 + ST + CNN-PIM$ and $CSP_6 + CWT + CNN-PIM$, the performance accuracy for the classification of two MIs improved significantly to

$97.62 \pm 02.31\%$ and $97.67 \pm 02.06\%$, respectively, with a low STD value.

The high STD values in Table 6 are high because as expected the intersubjects brain signals are different. Therefore, in BCI it is difficult to find a general model that works with the same performance for all the motor imageries registered by an electroencephalogram for all subjects. However, our proposed method $CSP_6 + CWT$ has an intrasubject STD low, and a better intersubject STD compared with state-of-the-art models. Furthermore, it is important to say that the MC-TM methodology was implemented with subjects that other works, like [11], discarded, contributing to the solution of BCI illiteracy. Therefore, the $CSP_6 + CWT$ feature extraction method was selected to be used in the methodology to classify four MIs.

3 Methodology to classify four MIs, MC-FM

The MC-FM methodology is an extension of the MC-TM methodology presented in Sect. 2. The datasets used for the MC-FM methodology of this Section are BCI-IV-2a described in Sect. 2A1 and a new dataset that is proposed in this paper to evaluate the MC-FM methodology with new subjects. In this case, the four MIs of the BCI-IV-2a dataset are used.

A. Our dataset

In order to make a more robust evaluation, a new dataset that includes the EEG signals of 15 available subjects was generated. Of the 15 subjects, 14 are men and 1 is a woman. The age range is from 22 to 27 years old. 2 men are left-handed (s_{14} and s_{15}), 2 men are ambidextrous (s_9 and s_{12}), and the woman (s_4) and the rest of the men are right-handed. Each subject signed an informed consent form. This form explained in detail the EEG sampling protocol and informed the subjects about the electrodes positioning and the confidential handling of the information obtained from their EEG signals. Then, a record was created for each subject. This record included the daily physical and mental conditions of the subject when the EEG sampling sessions were carried out. Also, all the inconvenience and anomalies presented during the EEG sampling sessions were recorded. The OpenBCI© cyton board, the USB-dongle, and the Ultracortex “Mark IV” EEG Headset with 8 AgCl dry electrodes were used to record the brain activity of each subject through an EEG. The sampling frequency used was $Fs_{our} = 250$ Hz. The 8 electrodes selected of the Ultracortex “Mark IV” EEG Headset correspond to

$$Ch = \{Cz, P3, C3, F3, AFz, F4, C4, P4\} \quad (17)$$

Table 3 Accuracies per subject of the proposed MC-TM methodology using the CNN-PIM with the validation set of EEGD dataset

S	CSP ₂	CWT	ST	CSP ₂ +C WT	CSP ₂ + ST	S	CSP ₂	CWT	ST	CSP ₂ +C WT	CSP ₂ + ST
1	79.25 %	84.75 %	77.25 %	94.00%	91.00%	28	85.75 %	88.25 %	93.50 %	94.00%	92.25%
2	81.25 %	89.25 %	91.75 %	96.25%	94.75%	29	74.25 %	92.75 %	93.75 %	91.75%	91.75%
3	91.00 %	88.00 %	89.50 %	96.25%	89.75%	30	81.25 %	93.25 %	92.75 %	90.50%	96.00%
4	87.75 %	86.50 %	81.50 %	97.75%	97.75%	31	72.00 %	83.00 %	94.25 %	75.75%	93.00%
5	90.25 %	92.75 %	94.25 %	84.00%	91.50%	32	67.00 %	87.50 %	90.25 %	87.50%	89.75%
6	80.25 %	93.25 %	88.00 %	88.00%	91.75%	33	65.50 %	85.00 %	90.75 %	89.25%	95.75%
7	84.25 %	88.50 %	88.50 %	87.25%	86.75%	34	80.25 %	93.75 %	77.00 %	91.75%	94.00%
8	89.00 %	79.75 %	89.50 %	86.75%	93.75%	35	86.50 %	93.25 %	92.75 %	93.75%	96.75%
9	76.50 %	83.75 %	91.75 %	86.50%	91.50%	36	83.50 %	90.50 %	89.00 %	97.50%	94.50%
10	78.00 %	87.00 %	92.25 %	98.50%	88.25%	37	76.25 %	92.75 %	92.25 %	96.25%	93.75%
11	75.25 %	81.75 %	90.75 %	92.25%	94.50%	38	74.75 %	84.75 %	89.25 %	91.50%	87.75%
12	80.00 %	75.25 %	86.50 %	90.25%	92.50%	39	73.00 %	57.25 %	68.25 %	93.25%	86.75%
13	80.25 %	81.25 %	90.75 %	96.00%	90.25%	40	76.50 %	71.25 %	81.00 %	92.00%	90.50%
14	79.25 %	99.25 %	96.75 %	90.00%	92.75%	41	93.75 %	78.75 %	82.00 %	87.00%	97.00%
15	92.75 %	89.50 %	93.50 %	98.50%	93.00%	42	79.75 %	89.00 %	91.75 %	92.50%	86.50%
16	84.25 %	86.00 %	93.75 %	91.50%	90.00%	43	93.75 %	97.50 %	87.25 %	99.25%	96.00%
17	79.00 %	90.75 %	91.50 %	93.25%	91.50%	44	78.00 %	81.25 %	87.00 %	91.00%	92.00%
18	73.75 %	83.75 %	86.25 %	95.00%	92.25%	45	72.50 %	69.75 %	69.00 %	90.50%	94.25%
19	72.50 %	94.25 %	92.50 %	92.75%	93.25%	46	61.75 %	73.25 %	65.75 %	83.50%	92.00%
20	83.00 %	77.75 %	90.00 %	92.25%	92.75%	47	79.50 %	71.25 %	87.50 %	89.00%	93.25%
21	81.50 %	80.75 %	87.25 %	90.75%	92.75%	48	84.00 %	74.50 %	83.75 %	97.25%	91.25%
22	85.25 %	86.75 %	91.25 %	93.75%	92.50%	49	83.00 %	89.50 %	88.75 %	81.75%	89.00%
23	87.50 %	90.50 %	87.75 %	93.25%	72.25%	50	77.50 %	96.50 %	91.75 %	89.00%	95.00%
24	73.25 %	91.25 %	89.00 %	91.25%	86.25%	51	77.75 %	75.25 %	88.50 %	95.00%	94.75%
25	88.75 %	89.25 %	89.50 %	93.00%	94.75%	52	87.50 %	88.50 %	91.75 %	85.75%	92.50%
26	81.75 %	69.50 %	81.00 %	93.25%	86.50%	AV G	80.61 %	85.06 %	87.91 %	91.59%	91.82%
27	91.00 %	83.75 %	89.75 %	94.00%	94.00%	ST D	7.08 %	8.36 %	6.58 %	4.56%	3.96%

Table 4 Accuracies per subject of the proposed MC-TM methodology using the CNN-PIM with CSP₄ and CSP₆ with the validation set of BCI-IV-2A dataset

S	CSP ₄ +C WT	CSP ₄ +ST	CSP ₆ +C WT	CSP ₆ +ST
01	88.72%	79.69 %	93.75%	93.06 %
02	79.86%	80.03 %	91.15%	90.10 %
03	95.31%	91.84 %	96.01%	99.48 %
04	85.59%	82.29 %	95.14%	88.02 %
05	84.72%	78.99 %	95.49%	82.64 %
06	79.86%	73.78 %	95.83%	92.53 %
07	80.56%	88.37 %	90.97%	84.72 %
08	93.92%	90.97 %	97.40%	93.58 %
09	90.45%	88.89 %	94.27%	93.75 %
AV G	86.55%	83.87 %	94.44%	90.88 %
ST D	5.92%	6.32%	2.18%	5.14%

according to the 10–20 international system. This electrode positioning is proposed using the Brodmann areas which include executive functions (F3, AFz, F4), motor functions (Cz), somatosensorial functions (C3, C4), and attention area (P3, P4). In addition, the A1 electrode was used as a reference electrode and the A2 electrode as a ground electrode [51]. Thus, the number of channels used for this dataset is $nCh_{our} = 8$.

A single trial of the EEG sampling protocol implemented in our dataset is presented in Fig. 4.

The proposed EEG sampling protocol is based on the Graz paradigm [52], which uses a synchronous system to acquire the EEG signals at a specific time. An advantage of using a synchronous system is the precise localization of the start and the end of an MI signal in the EEG signal. Before beginning the proposed EEG sampling protocol, the subject sits in a comfortable chair in front of a monitor, the Ultracortex Mark IV EEG Headset is positioned, and the EEG signal recorded is verified to be accepted. Then a beep sounds, $t = 0$, indicating the starting of a trial, and the screen presented in Fig. 5 is shown for 2 s to the subject

indicating a rest/break state. Later, at $t = 2$, a clue image of the MI that the subject must perform appears for 1.6 s. The possible MIs are

$$MI = \{LH, RH, FT, JC\} \quad (18)$$

The clue image is randomly selected, and it consists of an inanimate and opaque image that represents the MI. All the clues' images used in the EEG sampling protocol are illustrated in Fig. 6a. After $t = 3.6$, the clue image is animated, and it recalls the subject to perform the MI. The duration of the MI is 3.4 s. The clue implemented for the LH and RH is the one described in [11]. The clue illustrates the visualization of touching all the corresponding hand fingers with the thumb, starting from the pinky to the index finger. This action is repeated until the end of the trial. The clues are animated images to make more explicit the MIs to be executed. The clue for FT, was a person riding a bicycle, and in the JC clue an animation that shows the jaw clench (similarly to the chewing process). The MIs that the subject must execute are illustrated in Fig. 6b. The clues' images and the animations complement and reinforce the traditional Graz paradigm, with a more vivid visualization and stimulating the imagination of each of the subjects.

After $t = 7$, the subject receives a second rest/break state. This break is concatenated with the 2-s break issued at the starting of a new trial. Each session involves 12 trials of each MI. Therefore, each session has 111,001 samples, 96,000 of those samples correspond to the 48 trials (2,000 samples per trial). The other 15,001 samples correspond to the 60.004 s of brain activity where the subject was not focused on realizing an MI. In total, 12 sessions were registered by subject, i.e., $p_{our} = 144$. The processing of the EEG signals recorded with the proposed protocol, as well as the EEG signals obtained from BCI-IV-2a dataset, and the classifier for four MIs using the CSP₆ + CWT feature extraction method is described in the next sections.

B. EEG signal processing

The EEG signal processing stage is similar to the one implemented in Sect. 2B. The difference is that the following procedure is detailed using the EEG signals from our dataset instead of the EEG signals obtained from BCI-IV-2a dataset.

A specific sample of an EEG signal from our dataset is represented by

$$STR_{\Phi, \Delta, \Xi}^{Ch}(n); \Phi = 1, 2, \dots, 15; \Delta = 1, 2, \dots, 12; \Xi = 1, 2, \dots, 48; n = 1, 2, \dots, 2000 \quad (19)$$

where Φ represent the subject, Δ is the number of the session, and Ξ is the number of the trial.

All the samples of each of the trials of each of the sessions were chronologically concatenated, from the first

Table 5 Accuracies per subject of the proposed MC-TM methodology using the CNN-PIM with CSP₄ and CSP₆ with the validation set of EEGD dataset

S	CSP ₄ +C WT	CSP ₄ +S T	CSP ₆ +C WT	CSP ₆ +S T	S	CSP ₄ +C WT	CSP ₄ +S T	CSP ₆ +C WT	CSP ₆ +S T
1	96.25%	90.75%	94.75%	97.00%	28	95.75%	95.75%	94.50%	92.50%
2	98.00%	98.25%	98.00%	98.25%	29	97.00%	94.75%	97.50%	99.00%
3	97.25%	96.50%	100.00%	100.00%	30	98.75%	86.50%	90.25%	96.25%
4	99.75%	98.00%	99.75%	100.00%	31	98.75%	94.50%	99.75%	98.00%
5	97.00%	90.50%	95.50%	97.75%	32	90.75%	72.50%	98.75%	96.25%
6	97.00%	90.00%	97.50%	100.00%	33	98.00%	94.00%	97.75%	96.25%
7	94.25%	93.00%	97.50%	95.50%	34	98.50%	97.50%	98.00%	97.75%
8	91.00%	88.00%	97.00%	96.00%	35	99.50%	98.00%	98.25%	99.25%
9	98.25%	94.75%	97.25%	98.50%	36	98.75%	90.25%	99.00%	98.50%
10	100.00%	85.00%	95.00%	99.25%	37	98.75%	94.25%	100.00%	97.75%
11	92.00%	95.50%	96.50%	95.00%	38	99.50%	96.75%	98.00%	97.75%
12	88.75%	94.50%	93.00%	99.25%	39	98.50%	94.00%	98.50%	99.25%
13	99.25%	96.50%	98.50%	97.50%	40	98.25%	94.50%	97.75%	87.25%
14	100.00%	100.00%	100.00%	99.75%	41	97.50%	90.50%	100.00%	100.00%
15	97.75%	96.50%	99.00%	99.50%	42	95.75%	96.50%	96.75%	99.00%
16	95.75%	97.00%	99.00%	96.00%	43	98.00%	83.50%	94.75%	99.75%
17	99.00%	98.25%	97.75%	99.50%	44	95.00%	96.75%	98.00%	97.25%
18	97.75%	91.25%	99.00%	96.00%	45	98.00%	97.25%	96.75%	99.25%
19	98.75%	96.00%	98.50%	98.00%	46	97.25%	92.25%	98.50%	98.25%
20	99.25%	99.50%	97.00%	93.25%	47	97.75%	92.25%	98.75%	93.75%
21	98.25%	91.75%	97.25%	96.75%	48	99.75%	92.00%	98.75%	96.50%
22	96.25%	94.00%	98.25%	98.25%	49	96.50%	93.25%	92.25%	98.00%
23	95.50%	94.50%	100.00%	99.50%	50	98.25%	92.75%	99.00%	98.25%
24	94.75%	96.00%	99.50%	96.75%	51	98.25%	98.50%	97.75%	96.50%
25	98.75%	97.00%	100.00%	99.75%	52	99.50%	95.75%	96.50%	98.50%
26	99.00%	96.50%	97.75%	99.00%	AV G	97.26%	93.87%	97.67%	97.62%
27	95.75%	97.00%	99.75%	99.50%	ST D	2.42%	4.66%	2.06%	2.31%

sample of the first trial of the first session to the last sample of the last trial of the last session, to form a single signal.

$$\begin{aligned}
 Xraw_{\Phi}^{Ch}(n) &= STR_{\Phi, \Delta, \Xi}^{Ch}(n); n \\
 &= (n) + (\Xi - 1)(2,000) + (\Delta - 1)(96,000)
 \end{aligned}
 \quad (20)$$

here, $n \in \mathbb{N}^{1 \times \dot{N}}$, $\dot{N} = 1,152,000$ $Xraw_{\Phi}^{Ch}(n) \in \mathbb{R}^{n_{Chown} \times \dot{N}}$ is the discrete signal that contains the $\dot{N}$ raw samples of the EEG signal per channel of each subject. The EEG signal processing used in Sect. 2B was implemented applying the preprocessing, the epochs extraction, and the data augmentation stages to the $Xraw_{\Phi}^{Ch}(n)$ signal. The resulting

EEG signal processing is denoted by $x_{\Phi, ep}^{Ch}(k) \in \mathbb{R}^{nCh_{own} \times 2Fs_{own} \times 8p_{own}}$.

C. Feature extraction

According to the proposed methods of feature extraction analyzed in Sect. 2, the best method turned to be CSP₆ + CWT. Thus, this method was selected to be used in the classification of four MIs. To accomplish the four-MI classification, a modular classification scheme is proposed, and it will be analyzed in Sect. 3D. As a modular classification scheme is proposed, the spatial filter matrices were obtained using pairs of MI classes. For example, the first spatial filter matrix corresponds to LH and RH, the second to LH and FT, up to complete all the combinations {LH-RH, LH-FT, LH-JC, RH-FT, RH-JC, FT-JC}. As previously indicated, to simplify the mathematical notation the notation of the constants p , nCh , and Fs will omit the dataset and the subject Φ .

$Z_{CSP}^{ep, row}(k) \in \mathbb{R}^{6 \times 2Fs \times 8p}$ is obtained by using (3) and $row = \{1, 2, 3, nCh-2, nCh-1, nCh\}$ of the spatial filter $W_{CSP} \in \mathbb{R}^{nCh \times nCh}$ generated through the CSP for each pair of MI. Then, the CWT of $Z_{CSP}^{ep, row}(k)$ is calculated by (11). The result of the CWT is $Z_{CSP_CWT}^{ep, row, sc}(k) \in \mathbb{R}^{SC \times 2Fs \times 6 \times 8p}$. Then, the resulting matrices of the CSP + CWT method are used to compute the matrix

$$Q_{CSP_CWT}^{ep}(i, k) = Z_{CSP_CWT}^{ep, j, [i-(j-1)*SC]}(k); \quad (21)$$

$$Q_{CSP_CWT} \in \mathbb{R}^{6SC \times 2Fs \times 8p}$$

$$\text{where } i = 1, 2, \dots, 6SC; j = \begin{cases} 1 & 0 < i \leq SC \\ 2 & SC < i \leq 2SC \\ 3 & 2SC < i \leq 3SC \\ 4 & 3SC < i \leq 4SC \\ 5 & 4SC < i \leq 5SC \\ 6 & 5SC < i \leq 6SC \end{cases}$$

Finally, the bicubic interpolation is applied by using (13) to generate $I_{CSP_CWT} \in \mathbb{R}^{22 \times 100 \times 8p}$. I_{CSP_CWT} contains the input images for the CNN-PIM, with information of CSP₆ + CWT feature method. An example of four images of each MI included in I_{CSP_CWT} of the subject 03 of BCI-IV-2a dataset is illustrated in Fig. 7. It can be noticed that the signal behavior over time and frequency is consistent in the LH and JC MIs regarding the energy in the same MI. On the other hand, it is observed that it is complicated to visually discriminate the RH and FT MIs.

D. Four-MIs classifier

Considering the complexity to recognize four MIs, it was decided to face the classification problem from a

modular approach. This approach avoids designing a large and very complex classifier. Instead, smaller, and simpler classifiers are designed to solve simpler tasks. The classifiers are based on the CNN-PIM model shown in Fig. 3. A CNN-PIM was implemented to classify each pair of MIs. All the CNN-PIM were trained using the values given in Table 1. An illustration of the complete modular classification scheme is shown in Fig. 8.

In the modular classification scheme, the final predicted class is determined either using the mode or by an algorithm that was named secSM (selected class when compared the outputs of the SoftMax functions). First, the mode was used to determine the final predicted class by selecting the more repeated class determined by each of the CNN-PIMs. In practice, sometimes there was a tie between two classes that were determined by the CNN-PIMs the same number of times. In this case of tie, the final predicted class will be the class whose label number is the lowest. LH MI has the label “1”, RH has the label “2”, FT has the label “3”, and JC has the label “4”. The labels were assigned by the BCI-IV-2a dataset, and they were used as well in our dataset. In such a way that if there is a tie between two classes, e.g., FT and JC, then the selected class will be FT because it has the label with the lowest number.

To avoid the situation mentioned above, the proposed secSM algorithm evaluates if there is a tie between two classes. If there is a tie, a summation over all outputs of the softmax functions where the MI class was the determined class by each CNN-PIM is done to break the tie (in the next expression the CNN-PIM used are represented just by CNN for short)

$$\begin{aligned} sLH &= (\Psi_{CNN1}^{LH})(SM_{CNN1}^{LH}) + (\Psi_{CNN2}^{LH})(SM_{CNN2}^{LH}) \\ &\quad + (\Psi_{CNN3}^{LH})(SM_{CNN3}^{LH}) \\ sRH &= (\Psi_{CNN1}^{RH})(SM_{CNN1}^{RH}) + (\Psi_{CNN4}^{RH})(SM_{CNN4}^{RH}) \\ &\quad + (\Psi_{CNN5}^{RH})(SM_{CNN5}^{RH}) \\ sFT &= (\Psi_{CNN2}^{FT})(SM_{CNN2}^{FT}) + (\Psi_{CNN4}^{FT})(SM_{CNN4}^{FT}) \\ &\quad + (\Psi_{CNN6}^{FT})(SM_{CNN6}^{FT}) \\ sJC &= (\Psi_{CNN3}^{JC})(SM_{CNN3}^{JC}) + (\Psi_{CNN5}^{JC})(SM_{CNN5}^{JC}) \\ &\quad + (\Psi_{CNN6}^{JC})(SM_{CNN6}^{JC}) \end{aligned}$$

$$\Psi_{CNN-PIM}^C = \{0 : \text{non-predicted class}, 1 : \text{predicted class}\};$$

$$CNN - PIM = \{CNN1, CNN2, CNN3, CNN4, CNN5, CNN6\}$$

(22)

where $SM_{CNN-PIM}^{MI}$ are the outputs of the softmax function for each MI and each CNN-PIM, and $\Psi_{CNN-PIM}^C$ is 1 for the predicted class, C , and zero for the other class in each particular CNN-PIM. For example, if the predicted class of the CNN1 is the LH MI, then, $\Psi_{CNN1}^{LH} = 1$, $\Psi_{CNN1}^{RH} = 0$, and

Table 6 Accuracy comparison of the proposed feature extraction methods with other works reported in the literature, using the bci-iv and eeg datasets

BCI-IV DATASET							EEGD DATASET								
References	Feature extraction	Classifier	No. of MIs	No. of subjects	Performance	Dataset	Year	References	Feature extraction	Classifier	No. of MIs	No. of subjects	Performance	Dataset	Year
[19]	FFT—WPD	DL based on RBM	2	9	84.00 ± 11.93%	BCI-IV-2b	2017	[11]	CSP	FLDA	2	52	67.46 ± 13.17%	EEGd	2017
[15]	CSPO	TW-SVM	MIs		75.93 ± 15.43%	BCI-IV-2a		[36]	CSP	SVM	MIs		72.83 ± 08.48%		2020
[17]	CWT	CNN			78.93 ± 06.83%	BCI-IV-2b	2018	[46]	VMD + STFT	DeepConvNet			88.51 ± 10.64%		2021
[47]	AR-CSP	SRSG-FasAr			85.49 ± 11.64%	BCI-IV-2a				EEGNet			85.66 ± 07.49%		
[49]	TSGSP	SVM			82.51 ± 12.24%		2019	[48]	GFC				74.12 ± 12.13%		
[21]	Wavelet	CNN			85.29 ± 06.02%			Our	CSP ₂	CNN-PIM			80.61 ± 07.08%		-
[50]	JSTFD	LDA			79.60 ± 14.70%		2020	methods	CWT				85.06 ± 08.36%		
[36]	CSP	SVM			79.57 ± 10.89%				ST				87.91 ± 06.58%		
[48]	GFC				81.92 ± 09.44%		2021		CSP ₂ + CWT				91.59 ± 04.56%		
Our	CSP ₂	CNN-PIM			80.30 ± 13.72%		—		CSP ₂ + ST				91.82 ± 03.96%		
methods	CWT				76.41 ± 06.78%				CSP ₆ + ST				97.62 ± 02.31%		
	ST				69.77 ± 07.15%				CSP₆ + CWT				97.67 ± 02.06%		
	CSP ₂ + CWT				81.81 ± 11.34%										
	CSP ₂ + ST				79.19 ± 12.50%										
	CSP ₆ + ST				90.88 ± 05.14%										
	CSP₆ + CWT				94.44 ± 02.18%										

Bold means the method with better accuracy

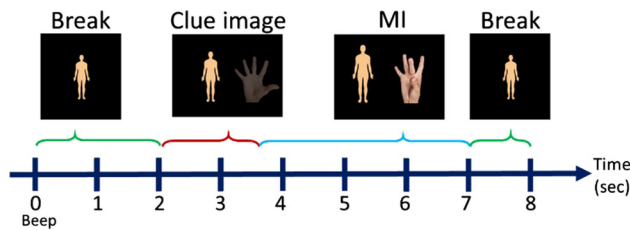


Fig. 4 An example of a trial of the proposed EEG sampling protocol

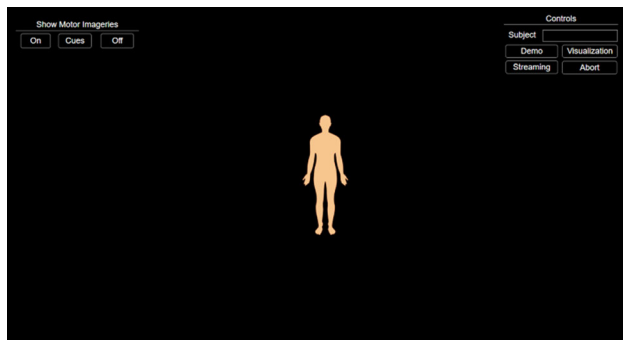


Fig. 5 Screen that is shown as a rest/break of the subject in the proposed EEG sampling protocol

if not $\Psi_{CNN1}^{LH} = 0$, $\Psi_{CNN1}^{RH} = 1$. Then, the final predicted class using secSM will be

$$out_{secSM} = \arg \max(sLH, sRH, sFT, sJC) \quad (23)$$

E. Results of the four-MIs classification

To train and validate the classifiers, a cross-validation with 5 folders was used in both datasets. The average accuracy for the cross-validation training sets for all subjects of the BCI-IV-2a dataset for each CNN-PIM was $97.62 \pm 1.07\%$ for LH-RH MIs, $98.02 \pm 1.1\%$ for LH-FT MIs, $97.98 \pm 1.92\%$ for LH-JC MIs, $97.85 \pm 2.3\%$ for RH-FT MIs, $97.46 \pm 1.55\%$ for RH-JC MIs, and $96.80 \pm 1.86\%$ for FT-JC MIs. On the other hand, the

average accuracy for the cross-validation training sets for all subjects of our dataset for each CNN-PIM was $92.93 \pm 3.6\%$ for LH-RH MIs, $91.09 \pm 4.39\%$ for LH-FT MIs, $90.03 \pm 5.84\%$ for LH-JC MIs, $91.34 \pm 3.78\%$ for RH-FT MIs, $89.45 \pm 7.58\%$ for RH-JC MIs, and $92.41 \pm 4.38\%$ for FT-JC MIs. A more complete analysis per subject for the average accuracies of the validation sets of both datasets is shown in Table 7. In Table 7 the accuracy was sorted in descending order according to the AVG column, and the color code was implemented by column by each type of classifier.

To implement the modular classification scheme, the parameters of each CNN-PIM were selected according to the best performance achieved in the cross-validation. Then, to examine the MC-FM methodology, the ROC curves of each pair of MIs classifiers were computed using a new test set. According to the availability of the data, the test set of the BCI-IV-2a dataset includes 216 samples per MI per subject, and the test set of our dataset includes 432 samples per MI per subject. The resulting ROC curves when all subjects were evaluated are shown in Fig. 9. As it can be seen in Fig. 9a, the behavior of the ROC curves for the BCI-IV-2a dataset is similar between them and with excellent True Positive and False Positive ratios. In terms of the AUC metric (Area Under the ROC Curve), the classifiers achieved 0.9871 for LH-RH MIs, 0.99 for LH-FT MIs, 0.9936 for LH-JC MIs, 0.9897 for RH-FT MIs, 0.9918 for RH-JC MIs, and 0.9892 for FT-JC MIs. In Fig. 9b, where our dataset is evaluated, the classifiers achieved an AUC metric of 0.9633 for LH-RH MIs, 0.9788 for LH-FT MIs, 0.9728 for LH-JC MIs, 0.9748 for RH-FT MIs, 0.9753 for RH-JC MIs, and 0.9644 for FT-JC MIs.

To complement the analysis of the MC-FM methodology, we also generate the confusion matrices of the proposed modular classification scheme that includes four MIs when the secSM algorithm was used and all subjects were evaluated, Fig. 10.



Fig. 6 Images shown in the proposed EEG sampling protocol for **a** Clues' images **b** Clues' images animated

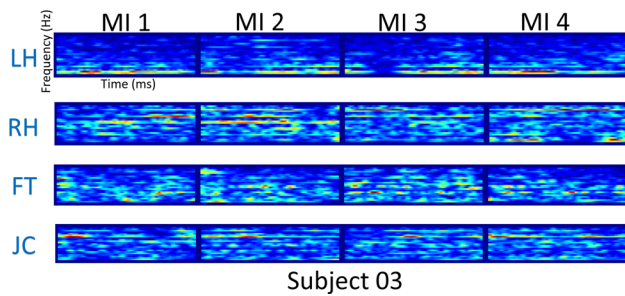


Fig. 7 Examples of $I_{\text{CSP-CWT}}$ of some MI signals of subject 03 of the BCI-IV-2a dataset

In Fig. 10, the sensitivity or recall metrics are presented in the squares with a light gray background and green color font on the right side of each confusion matrix. The precision metric is presented in the squares with a light gray background and green color font at bottom of each confusion matrix. The classification error is presented in the square with a dark gray background and red color font of each confusion matrix.

Then, the accuracies and the kappa coefficients per subject of the MC-FM methodology were computed Table 8 and sorted using the color code.

Table 9 shows the comparison between the proposed methodology to classify four MIs against the works [6, 21, 23, 24, 26, 47, 53–55], using the BCI-IV-2a dataset. It can be noticed that the accuracy of the proposed methodology in this paper achieves an average accuracy that exceeds the rest of the accuracies of the state-of-the-art methods reported when the BCI-IV-2a dataset was used. In addition, the proposed methodology attains an $\text{STD} = 3.2\%$ which implies a low dispersion of the accuracies in the classification.

The method VMD + STFT + EEGNet [46] shows the highest accuracy of $94.4 \pm 2.8\%$. However, this method uses EEG signals of 5 s, and it cannot be implemented in a real-time application. In addition, in Table 6 this method reports an accuracy of $85.66 \pm 07.49\%$ when the EEGd

dataset was evaluated. This is an inconsistency of the method because the EEGd dataset evaluates 2 MIs of 52 subjects and the BCI-IV-2a dataset evaluates 4 MIs of 9 subjects, and this method reports the highest AVG accuracy when 4 MIs are evaluated. Also, the intersubject accuracies produce a high STD value of 7.49% in the evaluation of the EEGd dataset. Instead, our methodology presents consistency because in the evaluation of the EEGd dataset it achieved an accuracy of $97.67 \pm 02.06\%$, and in the evaluation of the BCI-IV-2a dataset an accuracy of $91.37 \pm 03.29\%$. In both datasets, the intersubject accuracy values produce low STD values. Besides, our methodology uses EEG signals of 2 s with an approximate time-processing of 270 ms, which is suitable to be implemented in a real-time MI-based BCI system.

The kappa coefficient is also employed to compare the results of the proposed MC-FM methodology against the works reported in the literature. Only the works [23, 47, 53, 55] of the previous comparison (Table 9) presented the kappa coefficient metric in their papers. Therefore, to generate a more complete evaluation of our proposed methodology using the kappa coefficient, the works [56–59] were used too. The performances achieved in terms of the kappa coefficient are shown in Table 10. As it can be observed, the results of the proposed methodology exceed the rest of the performances by above 0.13 in terms of the kappa coefficient. Furthermore, the work described in [57] is the winner of the BCI-IV-2a dataset, and the proposed methodology outperforms it by 0.3.

To conclude this section, and to provide a more complete comparison, Table 11 summarizes a comparison between the performance of the proposed MC-FM methodology with the works compared in Table 9 (including more comparison elements) and other performances of the state-of-the-art works. The elements used in this comparison are: year of publication, number of MIs classified, types of MIs, number of subjects in the database, number of electrodes used in the EEG recording, the

Fig. 8 Proposed modular classification scheme to classify 4 MIs

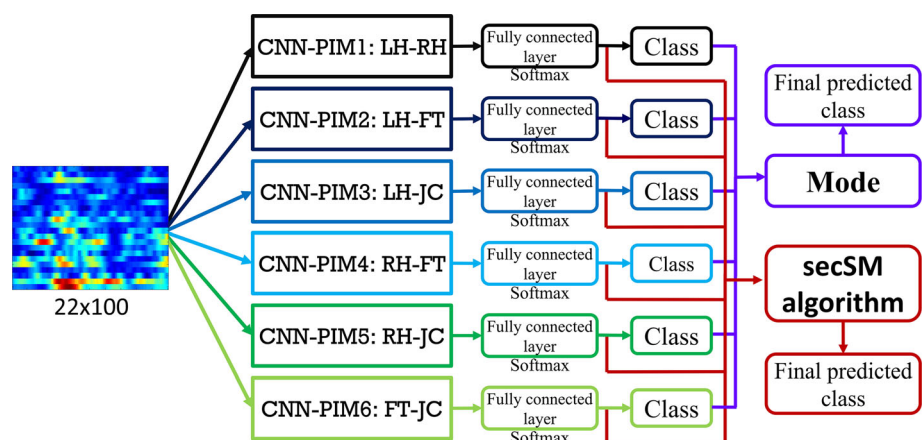


Table 7 Average accuracies of the CNN-PIMS per subject using the validation sets of both datasets: BCI-IV-2A and our dataset

BCI-IV-2A DATASET								OUR DATASET							
Subj ect / MI	LH- RH	LH- FT	LH- JC	RH- FT	RH- JC	FT- JC	AV G	Subj ect / MI	LH- RH	LH- FT	LH- JC	RH- FT	RH- JC	FT- JC	AV G
03	98.6 1%	98.4 7%	98.1 9%	98.7 5%	95.9 0%	93.8 2%	97.2 9%	S ₁₀	89.9 0%	95.3 0%	88.7 0%	96.3 0%	94.8 0%	89.2 0%	91.5 4%
01	96.0 1%	94.5 1%	98.9 6%	99.2 0%	99.8 6%	93.6 5%	97.0 3%	S ₈	88.8 0%	88.4 0%	94.6 0%	91.7 0%	94.3 0%	89.5 0%	91.3 7%
07	92.4 3%	98.6 1%	96.9 8%	98.6 1%	97.5 3%	91.6 7%	95.9 7%	S ₁	88.2 0%	92.8 0%	93.2 0%	90.0 0%	94.6 0%	90.0 0%	90.8 9%
05	94.5 8%	94.7 6%	95.9 0%	93.0 2%	95.1 7%	96.6 0%	95.0 1%	S ₁₁	90.7 0%	85.4 0%	93.4 0%	90.1 0%	87.2 0%	90.0 0%	88.1 4%
08	94.6 9%	94.1 0%	98.1 6%	94.5 5%	93.1 3%	95.0 3%	94.9 4%	S ₁₄	82.3 0%	86.8 0%	89.7 0%	87.2 0%	85.7 0%	88.7 0%	88.0 6%
09	93.7 5%	96.0 1%	97.6 7%	93.5 1%	93.6 5%	92.7 1%	94.5 5%	S ₃	90.2 0%	79.5 0%	85.3 0%	84.0 0%	90.6 0%	85.7 0%	86.6 1%
04	94.9 0%	95.4 5%	95.1 0%	93.8 9%	94.3 8%	92.7 4%	94.4 1%	S ₂	89.1 0%	87.4 0%	88.2 0%	84.8 0%	90.9 0%	90.7 0%	86.4 1%
02	90.0 0%	94.0 6%	91.6 0%	95.4 2%	92.4 3%	96.9 8%	93.4 1%	S ₉	89.8 0%	83.0 0%	83.6 0%	87.0 0%	84.6 0%	87.2 0%	85.8 1%
06	93.7 2%	92.0 5%	87.7 8%	87.6 4%	91.4 6%	88.8 5%	90.2 5%	S ₁₃	78.8 0%	83.4 0%	84.6 0%	87.2 0%	82.2 0%	83.2 0%	83.5 7%
AV G	94.3 0%	95.3 4%	95.5 9%	94.9 5%	94.8 3%	93.5 6%	94.7 6%	S ₇	79.2 0%	95.2 0%	93.3 0%	85.5 0%	79.5 0%	71.2 0%	82.9 4%
ST D	2.36 %	2.12 %	3.68 %	3.65 %	2.63 %	2.51 %	2.83 %	S ₁₂	92.0 0%	76.8 0%	73.1 0%	87.7 0%	83.7 0%	84.2 0%	81.5 4%
								S ₄	89.5 0%	84.3 0%	77.4 0%	79.2 0%	70.0 0%	85.6 0%	80.5 7%
								S ₁₅	81.5 0%	82.2 0%	74.9 0%	83.0 0%	84.3 0%	77.1 0%	79.4 4%
								S ₅	76.0 0%	84.9 0%	78.2 0%	81.8 0%	66.9 0%	80.2 0%	79.2 6%
								S ₆	84.1 0%	81.9 0%	83.5 0%	74.4 0%	78.9 0%	82.0 0%	78.7 3%
							AV G		86.0 1%	85.8 2%	85.4 5%	85.9 9%	84.5 5%	84.9 7%	84.9 9%
							STD		5.00 %	5.19 %	6.80 %	5.11 %	8.06 %	5.34 %	6.24 %

dataset used to evaluate the methodologies, approximated processing time (in seconds) in which the methodology has to determine the MI class, and accuracy. The results are sorted in descending order. As it can be observed in Table 11, most of the works use neural networks, and other methods use Support Vector Machine (SVM) or fuzzy systems. Most of the reported papers are evaluated with the BCI-IV-2a dataset. The work [27] is evaluated with BCI Competition II dataset III which includes the EEG signal of just one subject. It is well known that using the data of one subject does not represent a statistical significance. However, the work [27] was included because.

it shows the processing time in the classification and this value can help as a reference to compare other works. The

work [15] has the lowest classification time of 29.6 ms, and it uses spatial patterns and SVM to classify 2 MIs classes. This work achieved an accuracy of 75.93%. However, it distinguishes between 2 classes and its performance is under the performance of the proposed MC-FM methodology around 15%. It is important to say that the proposed MC-FM methodology uses 2-s windows to extract the MI event. Thus, as the proposed MC-FM methodology takes 270 ms to predict the output of the modular classification scheme, no MI is lost and it can be implemented in a real-time system, like the methods [15, 23, 60, 61]. In contrast, the works [6, 27, 49, 50, 55] have processing times over 2 s, and in consequence, these works cannot be seen as a real-time system. The method with the best reported

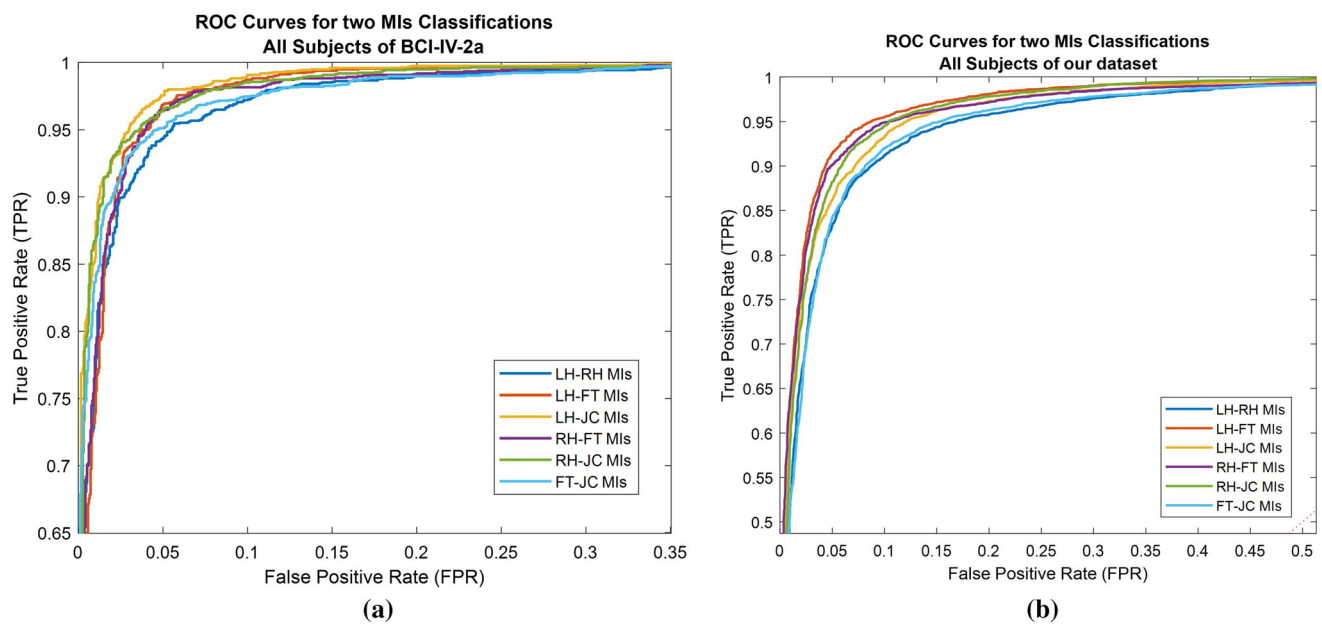


Fig. 9 Zoom view of the ROC curves of each pair of MIs when all subjects were evaluated using the test set of: **a** BCI-IV-2a dataset, **b** our dataset

LH-RH-FT-JC MIs from all subjects of BCI-IV-2a using secSM

Confusion Matrix					
Output Class	FT	JC	LH	RH	
	1726 22.2%	51 0.7%	36 0.5%	28 0.4%	93.8% 6.2%
	84 1.1%	1798 23.1%	32 0.4%	46 0.6%	91.7% 8.3%
	53 0.7%	61 0.8%	1809 23.3%	98 1.3%	89.5% 10.5%
	81 1.0%	34 0.4%	67 0.9%	1772 22.8%	90.7% 9.3%
Target Class					
	88.8% 11.2%	92.5% 7.5%	93.1% 6.9%	91.2% 8.8%	91.4% 8.6%

(a)

LH-RH-FT-JC MIs from all subjects of our dataset using secSM

Confusion Matrix					
Output Class	FT	JC	LH	RH	
	5552 21.4%	455 1.8%	155 0.6%	186 0.7%	87.5% 12.5%
	334 1.3%	5390 20.8%	126 0.5%	117 0.5%	90.3% 9.7%
	329 1.3%	376 1.5%	5772 22.3%	475 1.8%	83.0% 17.0%
	265 1.0%	259 1.0%	427 1.6%	5702 22.0%	85.7% 14.3%
Target Class					
	85.7% 14.3%	83.2% 16.8%	89.1% 10.9%	88.0% 12.0%	86.5% 13.5%

(b)

Fig. 10 Confusion matrices of the modular classification scheme that includes four MIs when the secSM algorithm was used and all subjects were evaluated using the test set of: **a** BCI-IV-2a dataset, **b** Our dataset

accuracy of $94.41 \pm 2.84\%$ was VMD + STFT + EEGNet [46]. Nevertheless, this method requires of 5 s windows, and it is not suitable for real-time applications.

On the other hand, the work [31] classifies 5 classes (4 MIs and an idle state, IS). However, its performance is 11%

lower than the proposed methodology. A more complete evaluation of the proposed methodology with our own dataset, including more subjects, shows a competitive result of $86.48 \pm 04.74\%$.

Table 8 Accuracies per subject of the modular classification scheme that includes 4 MIS using the test set of both datasets: BCI-IV-2A and our dataset

BCI-IV-2A DATASET				OUR DATASET			
Subject	Mode	secSM	Kappa of secSM	Subject	Mode	secSM	Kappa of secSM
01	95.25%	95.14%	0.9352	S ₁₀	92.13%	93.63%	0.9151
08	93.52%	94.91%	0.9321	S ₈	90.05%	92.65%	0.9020
03	94.44%	94.79%	0.9306	S ₁	89.81%	91.84%	0.8912
07	91.55%	92.48%	0.8997	S ₁₁	89.81%	91.61%	0.8881
04	88.43%	90.86%	0.8781	S ₂	87.04%	89.58%	0.8611
05	88.54%	90.63%	0.875	S ₉	84.32%	88.31%	0.8441
02	88.31%	89.93%	0.8657	S ₁₄	85.13%	87.73%	0.8364
09	85.42%	87.85%	0.8380	S ₃	81.83%	86.00%	0.8133
06	82.87%	85.76%	0.8102	S ₇	82.29%	84.43%	0.7924
AVG	89.81%	91.37%	0.8850	S ₁₃	80.67%	84.43%	0.7924
STD	4.20%	3.29%	0.0438	S ₁₂	77.78%	84.38%	0.7917
				S ₁₅	76.39%	81.94%	0.7593
				S ₄	78.65%	81.48%	0.7531
				S ₅	76.45%	80.67%	0.7423
				S ₆	72.40%	78.53%	0.7137
				AVG	82.98%	86.48%	0.8197
				STD	5.98%	4.74%	0.0632

4 Conclusions

This work aimed to develop a simple and efficient methodology to classify four MIs signals. To generate this methodology, named as MC-FM methodology, the work presents an analysis of five different feature extraction methods termed: CSP, CWT, ST, CSP + CWT, and CSP + ST. The feature extractor methods were evaluated using two datasets: BCI Competition IV dataset 2a and EEGdataset. The analysis carried out shows that using 6 rows in the CSP in combination with CWT is the best feature extractor method for the evaluated datasets, achieving validation accuracies of $94.44 \pm 02.18\%$ and $97.67 \pm 02.06\%$ in the classification of right-hand and left-hand MIs, for the BCI Competition IV dataset 2a and EEGdataset, respectively. Thus, the CSP₆ + CWT method was extended to classify four MI classes.

A classifier based on a CNN-PIM to classify two MIs, right-hand and left-hand was developed first to evaluate the feature extraction methods. Then a modular classification

scheme, that uses 6 instances of CNN-PIM, is used to classify four MIs, left-hand, right-hand, feet, and tongue motions. Each CNN-PIM is composed of 3 convolutional layers. Furthermore, an algorithm called secSM is proposed to improve the classification accuracy in the modular classification scheme, by using the outputs of the softmax functions of the CNN-PIMs. Another important contribution in this work, is the proposed EEG sampling protocol that helps to stimulate the subject's imagination and, in consequence, the generation of the visualizations. It also limits the number of motor imageries that the subject must realize, which contributes to the improvement of MI-based BCI systems because of its data-centric approach.

The results obtained from the proposed MC-FM methodology indicate that it can be used to acquire consistent MIs EEG signals from different subjects. It can include subjects who may have been discarded because their signals were not suitable for discrimination. Therefore, our work contributes to the solution of BCI illiteracy.

Table 9 Accuracies (%) comparison between the proposed CSP₆ + CWT + CNN-PIM method and other works reported in the literature evaluated with BCI-IV-2A dataset that includes 4 MI

Subject	CS + FLS + SO [6]	FB-CSP + CNN [26]	FBS + CSP + BO + CNN [23]	TSGL-EEGNet + stacking [53]	DFL + CD-Ioss [54]	AR-CSP + SRSG-FasArt [47]	PSD + CNN [24]	Ensemble rand sub. K-NN [55]	CSP ₆ + CWT + CNN-PIM	VMD + STFT + EEGNet [46]
01	81.4	85.2	84.9	85.4	91.3	95.1	86.0	98.9	95.1	93.5
02	67.5	69.7	66.3	70.6	71.6	68.7	87.0	97.6	89.9	97.7
03	88.2	90.1	84.7	95.2	92.3	98.6	93.0	97.6	94.7	93.6
04	48.6	65.5	81.3	80.2	78.3	81.2	85.0	93.8	90.8	91.0
05	52.9	77.4	79.2	70.2	80.1	72.2	84.0	80.2	90.6	94.8
06	55.7	52.4	70.6	68.3	61.6	75.0	86.0	88.6	85.7	95.2
07	84.3	93.6	86.1	90.9	92.6	86.1	88.0	71.2	92.4	96.4
08	89.3	90.0	83.8	86.3	90.3	97.9	89.0	94.8	94.9	89.4
09	85.7	84.7	83.0	84.4	78.3	94.4	92.0	90.3	87.8	97.6
AVG	72.6	78.7	80.0	81.3	81.8	85.4	87.7	90.3	91.3	94.4
STD	6.8	13.7	6.9	9.6	10.1	11.6	3.0	9.2	3.2	2.8

Bold means the method with better accuracy

Table 10 Kappa coefficient comparison between the proposed CSP₆ + CWT + CNN-PIM method and other works reported in the literature evaluated with BCI-IV-2A dataset that includes 4 MI

Subject	CR-SSELM [56]	FBCSP + OVR [57]	CSP + PWKNN [58]	FBS + CSP + BO + CNN [23]	AR-CSP + SRSG-FasArt [47]	CBN + SVM [59]	Ensemble random sub. K-NN [55]	TSGL-EEGNet + stacking [53]	CSP ₆ + CWT + CNN-PIM
01	0.71	0.68	0.88	0.68	0.79	0.69	0.97	0.805	0.935
02	0.38	0.42	0.22	0.36	0.42	0.51	0.93	0.609	0.865
03	0.70	0.75	0.88	0.69	0.79	0.87	0.93	0.936	0.930
04	0.49	0.48	0.39	0.62	0.63	0.85	0.83	0.736	0.878
05	0.26	0.40	0.53	0.6	0.42	0.78	0.47	0.604	0.875
06	0.32	0.27	0.33	0.45	0.34	0.42	0.69	0.578	0.810
07	0.76	0.77	0.38	0.71	0.84	0.54	0.23	0.879	0.899
08	0.79	0.75	0.85	0.72	0.65	0.97	0.86	0.818	0.932
09	0.73	0.61	0.81	0.66	0.78	0.45	0.74	0.793	0.838
AVG	0.57	0.57	0.59	0.61	0.63	0.68	0.74	0.75	0.88
STD	0.19	0.18	0.27	0.12	0.19	0.20	0.25	0.12	0.04

Bold means the method with better accuracy

Table 11 Accuracy comparison between the performance of the proposed methodology with other state-of-the-art methods

References	Year	No. of MI	Types of MI	Subjects	<i>nCh</i>	Dataset	Method	Approx. processing time (sec)	Accuracy
[46] Our	2021 –	4	Left, right, feet, ton	9	22	A ^a	VMD + STFT + EEGNet CSP₆ + CWT + CNN-PIM	– 0.270	94.41 ± 2.84% 91.37 ± 03.29%
[55]	2021						Ensemble of random subspace <i>k</i> -NN	2.89 ± 0.22	90.32 ± 09.24%
[24] Our	2018 –			15	8	C ^c	PSD (Welch) + CNN CSP₆ + CWT + CNN-PIM	– 0.270	87.78 ± 03.07% 86.48 ± 04.74%
[47] [21] [27] [49]	2018 2019	2	Left–right	9	22	A ^a	AR-CSP + SRSG-FasAr Wavelet + CNN	– –	85.49 ± 11.64% 85.29 ± 06.02%
				1	3	B ^b	RCT + ANN	5.25	84.40%
			Left–right / feet-ton	9	22	A ^a	TSGSP + SVM	17	82.51 ± 12.24% / 83.97 ± 07.54%
[60]	2020	3	Left, Right, Feet	6	9	C ^c	FBCSP + PLV2 + MK-RVM	0.515	83.81%
[54] [53]	2021 2021	4	Left, right, feet, ton	9	22	A ^a	DFL + CD-loss TSGL-EEGNet + stacking	– –	81.85 ± 10.77% 81.34 ± 09.64%
[62]	2021	2	Left–right	8	30	C ^c	fgLASSO	–	81.09 ± 07.05%
[23] [26] [50]	2018 4 2020	4 + IS 4	Left, right, feet, ton	9	22	A ^a	FBS + CSP + BO + CNN FB-CSP + CNN JSTFD + LDA	1.04 – 17.11	80.03 ± 06.52% 78.78 ± 13.78% 79.60 ± 14.70%
[15] [6] [22] [61]	2017 2018 2019				25		CSP + CPSO + TWSVM CSP + FLS + PSO FFTEM + CNN Renyi Entropy + LDA + Multi-class	0.0296 65 – 0.1559	75.93 ± 15.43% 72.62 ± 06.80% 68.00 ± 02.00% 62.58%

Bold indicates our proposed method

^aBCI-IV-2a dataset

^bBCI Competition II dataset III

^cTheir own dataset

The proposed MC-FM methodology takes advantage of the particularities of the feature extraction methods used in the literature to generate the features CSP₆ + CWT. As a result, it is a complete and robust methodology that can be implemented in real-time IoT healthcare systems due to its low processing time of 270 ms. Furthermore, the performances attained with the proposed methodology obtained competitive testing accuracies of 91.37 ± 03.29% and 86.48 ± 04.74% when it was evaluated with BCI Competition IV dataset 2a and with the proposed dataset, respectively, considering state-of-the-art methods.

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Declarations

Conflict of interest The authors have no conflicts of interest to declare that are relevant to the content of this article, and they certify that the manuscript complies with the Ethical Rules applicable for this journal.

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